

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q19-1334

Meadow Lake Wind Farm Llc 100 kV
November 18, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1334. The Interconnection Customer proposes a 141.8 MW solar photovoltaic facility interconnecting at the Meadow Lake Wind Farm Llc 100 kV substation in the Illinois area. The request was received on November 12, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 494 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$6,520,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Beach Haven - Bloomingdale 100kV Line
Contingency: Meadow Lake Wind Farm Llc - Weed Jct 100kV
Loading: 164.7% (Rating: 182 MVA)
DFAX: 100.0%
Pre-project loading: 86.9%
2. Facility: East F - Weed Jct 100kV line
Contingency: Beach Haven - Meadow Lake Wind Farm Llc 100kV
Loading: 162.7% (Rating: 187 MVA)
DFAX: 100.0%
Pre-project loading: 86.9%
3. Facility: Beach Haven - Meadow Lake Wind Farm Llc 100kV L
Contingency: Meadow Lake Wind Farm Llc - Weed Junction 100kV
Loading: 158.4% (Rating: 198 MVA)
DFAX: 100.0%
Pre-project loading: 86.9%
4. Facility: Weed Jct - Meadow Lake Wind Farm Llc 100kV Line
Contingency: Beach Haven - Meadow Lake Wind Farm Llc 100kV
Loading: 158.4% (Rating: 198 MVA)
DFAX: 100.0%
Pre-project loading: 86.9%
5. Facility: Ucon Substation - Kenneth-Garfield Tap 100kV
Contingency: Beach Haven - Meadow Lake Wind Farm Llc 100kV L
Loading: 147.7% (Rating: 103 MVA)
DFAX: 58.1%
Pre-project loading: 67.8%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Beach Haven - Bloomingdale 100kV Line	Rebuild line with higher-capacity conductor	\$5,039,000	43% (\$2,163,000)
East F - Weed Jct 100kV line	Rebuild line with higher-capacity conductor	\$2,911,000	43% (\$1,250,000)
Beach Haven - Meadow Lake Wind Farm Llc 100kV L	Rebuild line with higher-capacity conductor	\$2,111,000	43% (\$906,000)
Weed Jct - Meadow Lake Wind Farm Llc 100kV Line	Rebuild line with higher-capacity conductor	\$2,924,000	43% (\$1,255,000)
Ucon Substation - Kenneth-Garfield Tap 100kV	Rebuild line with higher-capacity conductor	\$2,203,000	43% (\$946,000)
	Total allocated to project		\$6,520,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.